Code No: 153SB **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech II Year I Semester Examinations, April/May-2023
MATHEMATICS – IV
(Common to CE, EEE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) Define the limit of a complex function *f(z)*. [2]
b) State Cauchy-Reimann equations in polar form and check the validity C-R equations for the function $f(r,v) = re^{i\theta}$. [3]
c) Define the generalized Cauchy's integral formula. [2]
d) Define an isolated singularity and essential singularity and give one example for each. [3]
e) Explain the process of evaluating an improper integral. [2]
f) Write the properties of the bilinear transformation. [3]
g) Write the Dirichlet's conditions for the existance of Fourier transform. [2]
h) Define Fourier sine and cosine integrals. [3]
i) Classify the partial differential equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + 2\frac{\partial u}{\partial x} + 5u = 0$. [2]
j) Find one independent solution of $y^3 \frac{\partial Z}{\partial x} + x^2 \frac{\partial Z}{\partial y} = 0$ by method of separation of variables. [3]

PART – B

(50 Marks)

2.a) Show that the function $f(z) = \sqrt{|xy|}$ is not analytic at the origin even though CR equations are satisfied.
b) Determine the analytic function whose real part is $e^{2x}(x\cos 2y - y\sin 2y)$. [5+5]

OR

3.a) If *f(z)* is a regular function of z, prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)|f(z)|^2 = 4|f'(z)|^2$.
b) Prove that $u = x^2 - y^2 - 2xy + 2x + 3y$ is harmonic function. [5+5]

4.a) Evaluate $\int_0^{1+i} z^2 dz$ along $y=x^2$.
b) Find Taylor's series expansion of $f(z) = \frac{2z^3+1}{z^2+z}$ about the point $z = i$. [5+5]

OR

5. Find the residue of $f(z)=\dfrac{z^3}{(z-1)^4(z-2)(z-3)}$ at its poles. Also evaluate $\oint_C f(z)dz$ where C is the circle $|z|=\dfrac{1}{2}$. [10]

6. By integrating around a unit circle, evaluate $\int_0^{2\pi}\dfrac{\cos 3\theta}{5-4\cos\theta}d\theta$. [10]

OR

7. Define the bilinear transformation and hence find the bilinear transformation which maps the points *z = 0, -i, -1* into the points *w= i, 1, 0.* [10]

8. Obtain the Fourier series for $f(x)=x-x^2$ in $[-\pi,\pi]$ and $f(x+2\pi)=f(x)\,\forall\, x\in R$.

Deduce the value of $\dfrac{1}{1^2}-\dfrac{1}{2^2}+\dfrac{1}{3^2}-\dfrac{1}{4^2}+\ldots\ldots=\dfrac{\pi^2}{12}$ from that Fourier series. [10]

OR

9. Find Fourier Sine transform and Fourier Cosine transform of $2e^{-5x}+5e^{-2x}$. [10]

10. Solve the one-dimensional wave equation by the method of separation of variables and hence find all the possible solutions. [10]

OR

11. An insulated rod OA of length *l* with insulated sides, has initial temperature $u(x,0)=f(x)$ for $0\le x\le l$. Both the ends are insulated at *t=0.* Find the subsequent temperature distribution. [10]

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